Psychology of Aesthetics, Creativity, and the Arts

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Online First Publication, July 23, 2020. http://dx.doi.org/10.1037/aca0000319

CITATION

Dumas, D., Organisciak, P., & Doherty, M. (2020, July 23). Measuring Divergent Thinking Originality With Human Raters and Text-Mining Models: A Psychometric Comparison of Methods. *Psychology of Aesthetics, Creativity, and the Arts*. Advance online publication. http://dx.doi.org/10.1037/aca0000319

AMERICAN PSYCHOLOGICAL ASSOCIATION
DIVISION 10
Psychology of Aesthetics, Creativity, and the Arts

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ISSN: 1931-3896
http://dx.doi.org/10.1037/aca0000319

Measuring Divergent Thinking Originality With Human Raters and Text-Mining Models: A Psychometric Comparison of Methods

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Within creativity research, interest and capability in utilizing text-mining models to quantify the Originality of participant responses to Divergent Thinking tasks has risen sharply over the last decade, with many extant studies fruitfully using such methods to uncover substantive patterns among creativity-relevant constructs. However, no systematic psychometric investigation of the reliability and validity of human-rated Originality scores, and scores from various freely available text-mining systems, exists in the literature. Here we conduct such an investigation with the Alternate Uses Task. We demonstrate that, despite their inherent subjectivity, human-rated Originality scores displayed the highest reliability at both the composite and latent factor levels. However, the text-mining system GloVe 840B was highly capable of approximating human-rated scores both in its measurement properties and its correlations to various creativity-related criteria including ideational Fluency, Elaboration, Openness, Intellect, and self-reported Creative Activities. We conclude that, in conjunction with other salient indicators of creative potential, text-mining models (and especially the GloVe 840B system) are capable of supporting reliable and valid inferences about Divergent Thinking. We offer an open-access module for researchers to apply these methods to their own data via our laboratory website (https://openscoring.du.edu/).

Keywords: creativity, divergent thinking, psychometrics, reliability, text-mining models

Supplemental materials: http://dx.doi.org/10.1037/aca0000319.supp

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Divergent thinking (DT)—or the human mental ability to generate multiple original ideas in response to a given problem or prompt (Acar & Runco, 2019; Forthmann, Wilken, Doebler, & Holling, 2019)—has long been a centrally important construct under investigation within the creativity research literature (e.g., Hocevar, 1980). Today, DT tasks are by far the most utilized measures in creativity research (Plucker & Makel, 2010; Reiter-Palmon, Forthmann, & Barbot, 2019), with the Alternate Uses Task (AUT; Guilford, 1967; Hudson, 1968; Torrance, 1972) being the chief among them. Therefore, procedures and methods that are used to calculate participant scores from their AUT or other DT task responses are closely considered, and at times psychometrically examined, by creativity researchers (e.g., Dumas & Dunbar, 2014; Kuhn & Holling, 2009). However, despite the broad use of the AUT in creativity research, the extant psychometric understanding of participant scores from the AUT is simply not as developed as in other areas of psychology (e.g., clinical personality assessment; Marek & Ben-Porath, 2017), for a variety of reasons.

In our view, one principal reason why DT measurement is somewhat underdeveloped is because DT tasks fundamentally rely on open-ended participant responses, whereas the vast majority of psychometric modeling frameworks that can be used to evaluate the reliability or internal validity of scoring models were designed for close-ended assessment (e.g., item response theory; Lord, 2012). In fact, the AUT is not only open-ended in its response format (i.e., participants must verbally respond or write-in their responses rather than selecting a response-option) it is also ill-structured (i.e., participants can differ substantially on the number of responses they supply and the length of those responses): an assessment format that stumps most currently dominant psychometric scoring methods. One approach to the scoring of open-ended divergent thinking tasks that focuses on taking a relatively rapid snapshot of creativity is a subjective scoring procedure introduced by Silvia and colleagues (2008). However, the search for more objective performance measurement strategies for divergent thinking tasks—especially measurement strategies that can be

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This research was supported financially through a research seed-grant from the University of Denver's Morgridge College of Education. We thank Amanda Strickland, Megan Solberg, and Danielle Francisco Albuquerque Vasques for their critical assistance in coding the responses.

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used as part of an automatic scoring system—has been more elusive (Dumas & Runco, 2018).

Fortunately, relatively recent advances in text-mining methodology (e.g., Hirschberg & Manning, 2015) are offering some solutions for the measurement of psychological attributes from open-ended and ill-structured assessment data. With the advent of a relatively new interdisciplinary area of research that combines text-mining and psychometric methods—sometimes termed *computational psychometrics* (von Davier, 2017)—the reliable quantification of human mental attributes that are too complex to be assessed via close-ended test items (e.g., DT) is becoming more possible than ever. In the next sections of this article, we first briefly overview areas in which computational psychometrics has opened important doors in psychological research and applied testing contexts, and then move to a detailed review of the existing computational psychometric applications in the creativity research literature.

Computational Psychometrics for Open-Ended Measures

For decades, social scientists engaged in basic research have understood that a close-ended psychometric methodological perspective—in which participants are administered questionnaires or tests, and the answer-choices that participants select are used to calculate their score—has some serious limitations for the study of complex human thought and behavior (e.g., Messick, 1995). For example, many investigations have utilized rich sources of interview, observation, essay, or otherwise open-ended and ill-structured data, which is typically scored through the application of human-raters who judge the degree to which those data indicate the presence of a given psychological construct within a participant (e.g., Bråten, Ferguson, Strømsø, & Anmarkrud, 2014). However, the use of such human-raters is costly both in terms of time and money, and the reliability and validity of human-rated participant scores is subject to measurement error from the raters' implicit beliefs and biases. The subjective nature of such human-rated scores has often been criticized in the literature (Gwet, 2014), although, in the case of open-ended and ill-structured data sources, no alternative has historically existed.

However, in the last decade, major advances have been made in devising methods for the objective (i.e., not human-rated) and automatic (i.e., computer-generated) scoring of open-ended psychological data sources, at least those that are textual or verbal in format. These computational psychometric approaches are based on text-mining models that are trained on a massive corpus of extant text (e.g., a library of digitized books; Crossley, Dascalu, & McNamara, 2017) to represent the semantic meaning of words in the context in which they are used. For example, the most popular of these text-mining models in the psychological literature today is Latent Semantic Analysis (LSA; Landauer, Foltz, & Laham, 1998), which is a dimensionality reduction technique that seeks to reduce a sparse-matrix of document-term counts, where each document is represented by the counts of all the words in the model vocabulary, to a much smaller representation of documents as a few hundred features. This is done by finding co-occurrence patterns between all the words in a document. For example, rather than representing the words *dog*, *puppy*, and *canine* all as independent quantitative counts, an LSA trained model would learn their relationship and represent each of those words similarly across the latent dimensions of the model. This has the effect of accounting for similar words and synonyms. Even though the words in "the dog barked" and "the puppy barked" are different, the contextual difference in not great, and they will appropriately be understood as similar in an LSA representation of those statements (Günther, Dudschig, & Kaup, 2015). The substantive effect of this modeling of latent dimensions is that words have quantifiable distances between themselves in the trained model, and how close or far two words are from each other tends to align with humans' perceptions of whether words are semantically similar or dissimilar (Landauer & Dumais, 1997).

This relatively nuanced representation of the semantic structure of language through text-mining models has allowed psychometricians to begin to objectively and automatically quantify such complex psychological constructs as writing ability (i.e., via essay prompts and automatic essay scoring systems; Foltz, Streeter, Lochbaum, & Landauer, 2013), depression (i.e., through the textual analysis of patient interview data; Kjell, Kjell, Garcia, & Sikström, 2019), and collaborative ability (i.e., by examining the semantic structure of online chats in which participants collaborate to solve an abstract problem; He, von Davier, Greiff, Steinhauer, & Borysewicz, 2017). Indeed, all of these applications are beginning to become relatively large scale, with automatic essay scoring systems being deployed widely in standardized admissions testing, mental health related text-mining systems beginning to make their debut in clinics, and the text-based assessment of student collaborative skills being administered to students all around the world as part of the Program for International Student Assessment (PISA) tests.

Computational Psychometrics in Creativity Research

Since the theorizing of Mednick (1962), creativity has often been considered from the viewpoint of *associative distance*. From this perspective, the Originality of a given response is conceptualized as arising from its distal-relatedness from the context or prompt from which it arose, such that a response that would be typically associated with a given prompt or context would be considered to be low in terms of Originality, whereas a response that is more unusual within a given context would be considered to be more Original. Importantly, the theoretical conceptualization of Originality as the associative distance between a prompt and the response a participant generates can be operationalized (at least for verbal or textual DT responses such as from the AUT) as the *semantic distance* between a given response and the AUT prompt (e.g., "Brick") from which it arose (Beaty, Silvia, Nusbaum, Jauk, & Benedek, 2014). This theoretical pairing of the associative distance among ideas and the semantic distance among the words used to express these ideas (i.e., the responses) may not be perfectly one-to-one (cf. historic "thought and language" debates in psychology; cf., Piatelli-Palmarini, 1980), but it appears close enough to justify the application of computational psychometric approaches (e.g., LSA) to DT data to quantify the semantic distances among the prompts and responses.

Latent Semantic Analysis and Divergent Thinking

In the creativity research literature, the text-mining model that is most typically utilized is LSA (Acar & Runco, 2019). LSA has

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been an appropriate and useful method for the quantification of Originality because the process of training an LSA system is effective at preserving the linearity of the relations among words, so that the semantic distances between them are directly comparable by studying the factorization matrix of words by latent dimensions. Using these matrices, the latent dimensions in an LSA model can be used as coordinates in a geometrically represented space, and the cosine of the angle between the word-vectors can be interpreted as the semantic or associative distance among words (Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990). As an example with the AUT, if the prompt was "fork," the response "eat pasta" would result in a vector that has an acute angle with the vector for "fork." In Contrast, the response "conduct electricity" would result in a vector that has a wider angle from the initial prompt vector. Please see Figure 1 for a visualization of the geometric relations among AUT prompt and responses in LSA. To calculate the Originality scores for these AUT responses, the cosine of the angle would be calculated to represent the semantic distance between the prompt and response, and then that semantic distance would be subtracted from 1 to yield an Originality score for each response.

Following this general methodological paradigm, a number of studies both of the measurement-related functioning of LSA based Originality scores (e.g., Dumas & Runco, 2018; Forthmann, Oyebade, Ojo, Günther, & Holling, 2019; Heinen & Johnson, 2018; Prabhakaran, Green, & Gray, 2014), as well the application of LSA Originality scores to answering substantive research questions in the creativity research literature (e.g., Hass, 2017a; White & Shah, 2016) have appeared. As far as we are aware, Kevin Dunbar and his students and collaborators (e.g., Dumas & Dunbar, 2014; Forster & Dunbar, 2009; Green, Kraemer, Fugelsang, Gray, & Dunbar, 2012) were the first to recognize the rich possibilities in the application of LSA to DT and other cognitive tasks. As such,

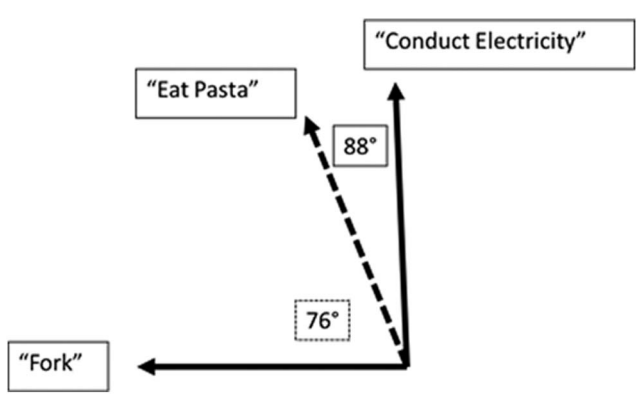

Figure 1. A visual representation of the geometric relations among vectors arising from an Latent Semantic Analysis (LSA) analysis. These angles were calculated using an LSA model trained on the Touchstone Applied Science Associates (TASA) corpus. To calculate the Originality of each of the Alternate Uses Task (AUT) responses, the cosine of the angle would be calculated to produce a semantic similarity score, and then that score would be subtracted from one to generate the Originality for each use. The cosine distance is taken to account for document length: Because words are represented in the latent model, multiword phrases can be represented as a sum of all word vectors (e.g., *vec(eat)* + *vec(pasta)*) while retaining the ability to compare them with one-word phrases.

Forster and Dunbar (2009) presented the first formal evidence of the appropriateness of LSA based Originality scores. In this work, they showed that LSA Originality scores were capable of discriminating between groups of participants who received different directions to the AUT (i.e., creative responses and common responses), and that LSA based Originality scores were more strongly predictive of human-rated Originality than were other common DT metrics such as Fluency and Elaboration.

Following this first foray into LSA based Originality scoring, Green and colleagues (e.g., Green, Kraemer, Fugelsang, Gray, & Dunbar, 2010) used LSA to examine the link between participant relational reasoning abilities and divergent thinking. In addition to this work, Dumas and Dunbar (2014) published the first psychometric investigation of LSA based Originality scores and their relation to Fluency scores from a latent variable perspective. This study found that a confirmatory factor analysis (CFA) model that included both LSA based Originality scores and Fluency counts across 10 AUT prompts was capable of fitting the observed data very closely and achieved a high degree of reliability for both the Fluency and Originality latent factors. In this way, LSA Originality scores have demonstrated discriminant validity from Fluency scores, and in another analysis, were shown to demonstrate a high degree of reliability even after the variance explained by Fluency was partialed out (Dumas & Runco, 2018).

Over the next few years, a solid handful of substantive applications of LSA Originality scores appeared in the creativity literature, demonstrating that this method was capable of producing scores that allowed creativity researchers to gain insights about psychological phenomena related to creativity. For example, White and Shah (2016) showed that LSA semantic distances among word association pairs were capable of explaining observed advantages of individuals with attention-deficit/hyperactivity disorder (ADHD) on a DT task, leading to the hypothesis that ADHD may support DT in part because of the wider scope of semantic activation in those individuals. The same year, Dumas and Dunbar (2016) showed that participants' LSA-based Originality scores were significantly influenced by DT task instructions, particularly when participants were asked to take the perspective of stereotypically creative individuals such as poets. The following year, two papers by Hass (2017a, 2017b) applied LSA-based Originality scores to a fine-grained analysis of DT production over short spans of time. For example, Hass (2017a) showed that the LSA-based semantic similarity (the inverse of Originality) of AUT responses were negatively correlated with human-judged creativity ratings, which provided evidence for the validity of LSA-based DT scores. In the same article, the semantic similarity of AUT responses followed a cubic trend overtime: The most highly semantically similar responses to the prompt occurred first, the similarity of responses then decreased (or Originality increased) over the first five or so responses, before increasing again between the fifth and 10th responses, and after the 10th response the similarity decreased once more. In addition, Hass (2017b) then demonstrated that the fluid intelligence of participants significantly influenced their ability to generate semantically distant (i.e., Original) responses to the AUT (although the change overtime of semantic distance was linear, not cubic in that investigation).

Later, Dumas (2018) used LSA-based Originality scores to test the long-standing threshold hypothesis (e.g., Karwowski & Gralewski, 2013) in the creativity literature: that intellectual ability

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supports creative ability, but only up to a point. In this study, LSA-based Originality scores on the AUT were able to reach a strong level of scale reliability, and allowed for an analysis in which the threshold hypothesis was supported under some conditions, but not in others. This finding, along with those of the Hass (2017a, 2017b) articles, may illustrate how semantic distance can be used as a fruitful operationalization of Originality to address longstanding questions in creativity research. In addition, Dumas and Strickland (2018) applied LSA-based Originality scores to investigate malevolent or violent responses on the AUT, and found that those participants who scored more highly on their Originality also produced more violent responses on the AUT (e.g., "kill someone" as a use for "shovel"). Such a finding adds to the potential evidence for the predictive validity of LSA-based Originality scores, because those scores were capable of significantly and positively predicting a theoretically relevant creativity-related construct (i.e., malevolence). Even more recently, Gray and colleagues (2019) used LSA-based scores to quantify the semantic distance among responses to word association tasks. Similarly to Hass (2017a), these researchers pointed out that the changes in semantic distance over time as participants generate responses is informative as to their creative potential.

Limitations of LSA

Since the very first applications of LSA in the psychological literature (see Landauer, McNamara, Dennis, & Kintsch, 2013 for a handbook of reviews), it has been understood that the reliability and validity of participant scores from computational psychometric models that incorporate LSA depend on a number of factors. Like all DT assessment, the quality of LSA-based Originality scores depends on a myriad of administrative and scoring choices, but some that are of particular importance in the LSA context include: (a) the corpus from which the LSA system was trained, (b) the particular methodological decisions for how to handle extremely common words like "and" or "is," and (c) the way in which semantic similarity or distance scores for individual DT responses are aggregated to the prompt (e.g., "Brick") level, or the participant level. Each of these issues is now briefly explained.

Training corpus. Beginning with Forster and Dunbar's (2009) initial work, by far the most widely utilized training corpus for LSA in the creativity research literature has been the Touchstone Applied Science Associates (TASA) corpus. This corpus was originally created by Landauer and Dumais (1997) and was initially applied to the psychological research literature by Walter Kintsch and his colleagues (e.g., Kintsch & Bowles, 2002). As part of the interdisciplinary work among these scholars, a freely accessible tool was made available to access this corpus, originally through an Internet browser (i.e., lsa.colorado.edu) but today also through the open-source software *r* (i.e., LSAfun; Günther et al., 2015). This corpus is composed of nearly 40 thousand educational texts and is meant to represent the average reading experience of the typical entering American undergraduate student (who are the most commonly recruited participants in psychology studies). However, the TASA corpus was created in the late 1990s and, as far as we are aware, has not been updated since the very early 2000s, calling into serious question the capacity of this corpus to continue to represent the true semantic relations of current language. Because the goal of the system is to adequately learn relations between words, more training texts generally lead to a better sense of the true semantic structure of language, and the TASA corpus is relatively small by modern standards. Finally, the type of texts trained on will affect the relations between words because, say, new articles, conversational posts, and legal arguments all have different styles of language. Which domain of corpora leads to semantic models that are most appropriate for Originality scoring is not yet known, and such a question seems worth exploring beyond the formal educational texts uses for the TSA corpus.

Currently, only a small minority of creativity researchers have used alternative training corpora for their LSA-based investigations. For example Forthmann and colleagues (2019) have used the more modern English 100k corpus, which is much more generally based on a web crawl of .uk domains of the Internet. However, most LSA-based creativity research (e.g., Dumas & Dunbar, 2014, 2016; Gray et al., 2019) continues to be done using the TASA corpus, raising a possible validity risk.

Elaboration confound. Because participant responses to DT tasks can contain varying amounts of words (i.e., they can vary in their Elaboration), LSA-based scoring may be adversely affected by these differences. As first pointed out by Forster and Dunbar (2009) and technically examined by Forthmann and colleagues (2019), LSA-based Originality scores at the participant level commonly exhibit a substantial correlation with Elaboration, implying that the more words a participant uses to explain their idea, the more LSA estimates of Originality are confounded. As a slightly more technical description, because LSA Originality scores are based on vectors for the entire DT response and not just individual words (see Figure 1), those vectors are essentially composed of the sum of the individual word vectors for every word in the DT response (Landauer, Laham, Rehder, & Schreiner, 1997). Because some words used in a response (e.g., "and" or "so") may be very commonly utilized, they are not particularly discriminatory. These commonly used function words lower the semantic distance of the full DT response from the prompt, even though the core idea of the response may have been highly Original. Such an understanding of LSA makes it clear that, going forward, DT tasks must be scored for Elaboration to identify (and possibly control for) confounds in substantive studies. In addition, the relation between text-mining–based Originality scores and human rated Originality scores needs to continue to be checked to ensure that the influence of Elaboration does not throw-off this relation. In addition, Forthmann and colleagues (2019) do offer some statistical corrections for formulating LSA-based Originality scores that can alleviate this problem. For instance, one way to control for the misleading effects of common function words is "stopword lists," which simply remove (or stop) a set of words based on a known list. In this article, a correction known as *term weighting* is applied to all systems under investigation, to weight different words to be more or less impactful based on how discriminatory they are. Here, Inverse-document-frequency (IDF) from the information science literature is used as the term weighting method (Robertson & Jones, 1976), using precomputed term weights that emphasize the influence of less common words (Organisciak, 2016). Correlations with Elaboration scores are also checked here as an index of discriminant validity.

Scoring aggregation method. One perhaps unfortunate pattern within the creativity research literature is that DT assessments are often thought of by researchers simply as tasks and not mea-

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sures. Although this is a subtle distinction, task is a much more general category that includes any stimuli designed to elicit a certain cognitive process or behavior from participants, whereas a measure requires multiple items or indicators of an underlying latent mental attribute to be aggregated, or scored, to represent a psychologically meaningful quantity (Hedge, Powell, & Sumner, 2018). For example, if the AUT is administered to participants, but only one or two prompts (e.g., "Book," "Hammer") are included, those few prompts cannot be aggregated in a way that provided psychometric evidence of the reliability and internal validity of the scores. In the creativity literature, observed-variable checks of the composite reliability (e.g., Cronbach's alpha) are not necessarily always done before using scores for analysis, and still rarer are studies of the underlying dimensionality and measurement properties of a set of DT items. Given that such formalized descriptions of the way DT prompts are aggregated into psychometric scores are rare in the creativity literature, it is difficult to build convincing arguments for the reliability and validity of any DT measure or scoring system, including the LSA-based Originality scores. For example, after generating LSA semantic distances for every response in the dataset, those responses are often averaged, or perhaps summed, within each prompt for every participant (see Forthmann, Szardenings, & Holling, 2020 for a close investigation into the effects of these methodological choices). Then, if multiple DT prompts (or specifically AUT prompts) were administered, participant scores across those prompts need to be aggregated in a reliable way (e.g., CFA) so that a score that represents participant level Originality can be produced. However, the psychometric properties of such a scoring model (if one is used) are rarely reported in the literature, limiting what is known about LSA-based Originality scores. Dumas and Dunbar (2014) were an exception to this, and they found relatively strong evidence for the psychometric reliability of LSA-based Originality scores, but they did not examine the relation between those scores and human-raters or Elaboration scores, among other limitations. It should be noted that the lack of strong psychometric evidence is a problem across much of the creativity research literature, not just LSA-based Originality scores, but the problem may be particularly poignant here, where the automated nature of these scores make the large-scale measurement of DT possible and provide opportunity for higher-stakes applications of creativity assessment, where low reliability can pose serious scientific and ethical problems.

Moving Beyond LSA in Creativity Research

Based simultaneously on the psychometric and psychological evidence in support of text-mining-based Originality scores, as well as more pragmatic and practical considerations such as the speed, cost, and objectivity of these methods, the continued use of text-mining models to score DT tasks seems justified and desirable. However, because of specific methodological concerns about LSA as a method generally, the TASA corpus specifically, as well as a general lack of formal psychometric investigation into these scoring systems, thinking critically about other possible text-mining systems for our work, beyond LSA TASA, appears important. Indeed, it may be that the free availability of the TASA-trained LSA tool—as well the fact that it was the text-mining system originally chosen by Forster and Dunbar (2009)—drives the creativity research literature's choice of this text-mining system over others that may provide us with better information about Originality.

Today, there are a number of other freely available text-mining systems in the information science and computer science communities that creativity researchers may recruit for their work. These available text-mining systems differ on the type of model they utilize (i.e., they do not use LSA), the text corpus they used to train the model, and the preprocessing and parameterization performed in training. For this reason, even across freely available text-mining systems, there is a high potential for very different psychometric and psychological patterns to emerge in Originality scores. For example, beyond the LSA models trained on the TASA and EN 100k corpora that have been used previously in creativity research, other text-mining systems are also freely available to researchers and could potentially be better for the quantification of Originality then LSA. One such system comprises the Google News model associated with the Word2Vec algorithm (Mikolov, Chen, Corrado, & Dean, 2013)—which is trained on 100 billion words scraped from the website Google News using a more modern neural network-based training approach. In addition, the Global Vectors for Word Representation (GloVe; Pennington, Socher, & Manning, 2014) algorithm provides a series of free, trained systems, including one based on 840 billion words scraped from across the Internet. The GloVe system, by virtue of its probabilistic (and therefore possibly more stable) statistical underpinnings and massive training corpus, may be more capable of producing reliable and valid Originality scores than previously used text-mining systems, although such a research question has never been systematically addressed. These models are described in more detail in the methodology section of this article, and are generally distributed as vector spaces providing a mapping of words to latent dimensions, which can be used programmatically to measure semantic distance.

As previously reviewed, a number of freely available text-mining models that have been trained on existing corpora of text and that are designed to represent the semantic structure of language through the estimation of word vectors within semantic space exist in the literature (i.e., TASA, EN100k, word2vec, GloVe). As will be delineated further in the Method section of this article, each of these text-mining models essentially utilizes a dimensionality reduction technique to quantify the relations among words or phrases by examining the angles among vectors (Landauer et al., 1997; see Figure 1). However, another technique exists within the psychological literature that offers an alternative to all of these dimensionality-reduction-based methods: the network modeling perspective (De Deyne, Verheyen, & Storms, 2016; Kenett, Levi, Anaki, & Faust, 2017). In this body of work, the semantic distance among words is not quantified by examining the angles among word vectors, but instead the length of a network path between two words (i.e., the number of intervening words in the network) is used as a measure of semantic distance. Recent work in cognitive psychology (Kumar, Balota, & Steyvers, 2019) has compared the network science approach to understanding semantic distance and dimension-reduction models LSA and word2vec, with some results showing advantageous properties of the network approach. In the current study, the possibility for incorporating network models into a computational psychometric approach to divergent thinking (see Kenett, 2019 for an overview)

is not investigated, although it is discussed later in this article as a future direction.

The Current Study

Given the current state of the creativity literature surrounding the quantification of Originality via text-mining models, coupled with the relatively recent availability of text-mining systems substantially more advanced than LSA and the TASA corpus, we have undertaken a systematic study of four major freely available text-mining systems (explained in more detail in the Method section) and the reliability and validity of the Originality scores they produce. These systems all rely on a different mix of methods, technical implementations, and training corpora, and we seek to understand which systems are more appropriate for scoring AUT Originality. Specifically, we aim to assess the internal consistency and factor reliability of AUT Originality scores produced by these text-mining systems at both the scale and latent-variable levels and compare that reliability to that of human-raters who judged the Originality of each AUT response. In addition, the predictive validity of the five Originality scoring systems (human raters and four different text-mining systems) is examined in terms of their correlation to a number of theoretically relevant DT dimensions (i.e., ideational Fluency and Elaboration), creative personality characteristics (i.e., Openness and Intellect) and self-reported real-world creative activities. The overarching goal of this investigation is to provide creativity researchers with psychometrically supported recommendations as to how to score DT responses for Originality, and what general predictive patterns to other creativity-related constructs (e.g., Elaboration) may be expected depending on what scoring system researchers choose to use.

Method

Participants

This study, which was part of a larger and ongoing investigation into the psychometrics of creativity, included 92 (53 female; 57.6%) participants. Participants were recruited for this study via Amazon Mechanical Turk, a crowdsourcing platform widely used in psychology research, including creativity research (e.g., McKay, Karwowski, & Kaufman, 2017). Because of the high language demands of divergent thinking tasks, participants were required to report themselves as fluent English speakers to participate, although two participants (2.1%) reported English as their second (but fluent) language. Participants were compensated $3.00 each for their participation. Participants were required to be over the age of 18 to participate, but the minimum actual participant age was 21, with a maximum age of 68. The mean age of participants was 37 (SD = 10.58). The majority of participants (n = 68; 73.91%) reported their race/ethnicity as European American, whereas smaller proportions of the sample reported their ethnicity as African American (n = 6; 6.5%), Asian (n = 9; 9.8%), Latinx (n = 5; 5.43) or multiple ethnicities (n = 4; 4.2%).

Measures and Tasks

Alternate uses task. The AUT is a psychometric measure in which participants are asked to generate as many creative uses for an object as possible within a certain amount of time (i.e., two minutes per object in this case). The AUT has been used for assessing divergent thinking and creative ability for decades (Guilford, 1967; Hudson, 1968; Torrance, 1972) and remains one of the most-often utilized tasks within the creativity research literature (e.g., Dumas & Strickland, 2018; Puryear, Kettler, & Rinn, 2017). The following 10 object names were presented to participants in a randomized order: *book, fork, table, hammer, pants, bottle, brick, tire, shovel,* and *shoe*. In this investigation, 10 AUT prompts (rather than a single AUT prompt as is often the case in creativity research) were used to reduce the stimuli dependence of the Originality scores (Barbot, 2018). This issue of stimuli dependence, and the concomitant need for multiple DT indicators, may be even more critically important when scoring the AUT with text-mining systems, because different AUT prompts (e.g., Book) may be represented in any given corpus differently, and therefore multiple stimuli are needed to produce the most reliable and valid scores. In this investigation, the 10 object names that were presented to participants were chosen to be in line with past work within the DT literature that has incorporated a text-mining approach (e.g., Dumas & Dunbar, 2014), as well as common practice within the DT assessment field, where objects are typically chosen that are expected to be familiar to participants, and that are reasonably different from one another to provide a reasonably broad sampling of object types, therefore reducing dependence on any one stimuli (Acar & Runco, 2019). Scoring procedures and resulting reliability and validity evidence for AUT scores are the main focus of this investigation, so that specific information is presented later in the Results section of this article.

Big Five Aspects Scale. The Big Five Aspects Scale (BFAS; DeYoung, Quilty, & Peterson, 2007) is a widely utilized self-report personality measure in which participants indicate levels of five principal aspects of personality, each of which is divided further into two facets. The "big five" dimensions of personality—Neuroticism, Agreeableness, Conscientiousness, Extraversion, and Openness—are all available on this measure, but the Openness scale is of particular interest to the present investigation, because this dimension of personality is the most perennially associated with divergent thinking, both in theory (Hornberg & Reiter-Palmon, 2017) and in empirical findings (Furnham, Crump, & Swami, 2009). The Openness dimension is further divided into two facets—Openness and Intellect—and both of these facets have been shown to be significantly and positively related to divergent thinking and creative outcomes, and are considered the core of the creative personality (Oleynick et al., 2017), making them both useful validity criteria in this study. In particular, we conceptualize the Openness and Intellect facets as providing important validity information in the following way: Intercorrelations among the text-mining-based Originality scores and the Openness and Intellect facets should, if the validity of the text-mining methods is upheld, be similar to the intercorrelations of the Openness and Intellect facets and the human-rater-based Originality scores. If Originality scores from one or multiple text-mining models were to display correlations with Openness and Intellect that were substantially different from human-judged Originality, the validity of that text-mining model would be called into question.

Although the most common method used to score self-report measures like the BFAS in psychology research is through summing the items, the summation of scores makes a number of strict

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measurement assumptions that are unlikely to hold (McNeish & Wolf, 2020). Therefore the Openness and Intellect scores for this study were generated through confirmatory factor analysis (CFA) by fitting a two-factor correlated model to both scales at once, as DeYoung and colleagues (2007) intended and validated. Specific psychometric information for each scale appears below.

Openness. The 10-item Openness facet of the BFAS has been particularly associated with creative outcomes, because it features such self-report items as *I need a creative outlet* and *I believe in the importance of art.* Participants indicated the degree to which each statement was true of them by dragging a 100-point slider with poles of 100 = *strongly agree* and 0 = *strongly disagree.* After reverse-coding all negatively worded items, in this study, the 10 items on the Openness facet of the BFAS achieved a scale internal consistency of $\alpha = .839$, with latent factor internal consistency indices based on factor loadings and uniquenesses being $H = .896$ and $\omega = .861$. Openness scores were generated from the CFA model via empirical Bayes and saved in the dataset.

Intellect. The intellect facet of the BFAS also contained 10 self-report items, including *I like to solve complex problems* and *I am quick to understand things*. Participants responded to these items in the same manner (i.e., with a slider) as they did the items on the Openness facet. After reverse-coding all negatively worded items the 10 items on the Intellect facet of the BFAS achieved a scale reliability of $\alpha = .840$, with latent factor internal consistency indices based on factor loadings and uniquenesses being $H = .876$ and $\omega = .858$. Intellect scores were generated from the CFA model via empirical Bayes and saved in the dataset.

Inventory of Creative Activities and Achievements (ICAA). The ICAA is a relatively recently developed (i.e., Diedrich et al., 2018) self-report measure for real-life creative activities and accomplishments across eight domains: literature, music, arts and crafts, cooking, sports, visual arts, performing arts, and science and engineering. Given the general nature of this sample, and time-constraints on the data collection, we administered the creative activity scale (rather than achievement) for six of those original eight domains: music, literature, arts and crafts, cooking, visual arts, and performing arts. Each of these scales consisted of six Likert-style items that ask participants how many times they have done particular creative activities in the past 10 years with five response categories: *never*, *1–2 times*, *3–5 times*, *6–10 times*, and *more than 10 times*.

For example, in the music domain, participants are asked how many times they have written a piece of music, or created a mix tape, among other items. In the arts and crafts domain, participants are asked how many times they created an original decoration. In cooking, how many times they made up a new recipe. The visual and performing arts scale asks how many times participants painted a picture and performed in a play, respectively. In this sample, each of the scales of the ICAA achieved satisfactory scale reliability as well as satisfactory latent factor reliability based on scale-specific single-factor CFA models, with literary activities having the lowest reliability ($\alpha = .800$; $H = .837$; $\omega =. 814$), music activity having the highest ($\alpha = .909$; $H = .968$; $\omega = .915$), and the other scales (visual arts; $\alpha = .826$; $H = .908$; $\omega = .836$; cooking; $\alpha = .874$; $H =. 903$; $\omega = .876$; performing; $\alpha = .876$; $H =. 899$; $\omega =. 882$; crafts; $\alpha = .90$; $H =. 921$; $\omega =. 905$) being in the middle. Taken together, all 24 items on these six scales displayed a composite scale reliability of $\alpha = .926$. and, as a single latent factor, latent factor reliabilities based on loadings and uniquenesses of $H = .952$ and $\omega = .934$. For future analysis, empirical Bayes-based latent factor scores were computed for each of the six administered ICAA scales, as well as a total Creative Activity that incorporated all six scales.

In this study, the ICAA is included as a validity-criterion measure with which to correlate the Originality scores produced by both human raters and the various text-mining models. In general, we conceptualize this validity procedure as requiring the text-mining-based Originality scores to approximate, in their correlations to the ICAA, the nature of the human-rated Originality scores. This validity-criteria procedure is based on the general problem in creativity research that an ongoing need to utilize human raters in our work creates a bottleneck to scaling creativity research to very large data sets. Here, we seek to test the capability of the text-mining models to create Originality scores for the AUT that are similar to those produced by humans, but much more rapidly and at a much lower cost. Hence, the ICAA serves as an external validity criterion to ascertain whether the text-mining models are successful at accomplishing this goal.

Administration Procedures

All participation for this study was conducted online via Mechanical Turk, and the study website itself (which participants were provided a link to) was hosted by Qualtrics. Informed consent was obtained before participants could move forward with the measures (these procedures were approved by the institutional review board at the Institution where the study took place). Study instructions asked participants to complete the measures with minimal distractions and recommended that they turn off electronic devices as well as close other websites or programs open on their computer. Because the AUT requires a significant amount of typing, participation required a traditional keyboard and participation via smartphone or tablet was not allowed. Participants were given two minutes to provide uses for each object before they were automatically advanced to the next object, and they could not advance before those two minutes were up. After responding to all 10 objects (i.e., after 20 min), participants were informed that the task was complete, and moved to the self-report portion of the study. In this phase, participants first provided responses to the ICAA and then moved to the BFAS. Finally, participants responded to the demographic question and logged out of the study website.

AUT Scoring Procedures

The main focus of this investigation was to examine the reliability and criterion validity of multiple Originality measurement methodologies for the AUT. As such, the AUT was scored a number of different ways in this study, each of which is detailed below.

Fluency. As a criterion by which to examine the validity of Originality scoring methods, the AUT was scored for Fluency. First, the number of uses generated by each participant for each object was tallied, and then summed across all 10 items on the AUT, producing a "total-uses" variable for analysis. Counts such as these are the principal way in which fluency has been operationalized in the extant literature (Plucker & Makel, 2010). In this

investigation, fluency counts across the 10 items on the AUT exhibited a high level of scale internal consistency (α = .946). However, to avoid making potentially untenable measurement assumptions, Fluency scores were generated via empirical Bayes from a single factor CFA model fit the 10 Fluency indicators. The scale exhibited latent factor reliability indices of H = .962 and ω = .957.

Elaboration. Also following well-established scoring procedures in the divergent thinking literature (e.g., Forster & Dunbar, 2009; Torrance, 1988), participant Elaboration scores were calculated by averaging the number of words utilized per response within each of the AUT prompts. In this scoring procedure, averaging within the AUT prompt is meant to reduce the implicit association between Elaboration and Fluency (i.e., those participants who generated more responses will have used more words in total, but perhaps not on average). However, a statistical relation between these two dimensions of divergent thinking may still exist regardless of this scoring choice because of a possible psychological association between ideational fluency and the ability to elaborate on those ideas (Hudson, 1968). In addition, the strength of the relation between Elaboration and Originality has been the focus of previous investigations of text-mining scoring systems for Originality (Forthmann et al., 2019), and therefore it is of high importance here. These 10 AUT prompt elaboration scores displayed a high level of scale reliability (α = .958), and at the latent factor level (via a single factor CFA), displayed strong latent internal consistency indices H = .965 and ω = .961. Elaboration scores for each participant were generated via empirical Bayes from the single factor CFA model.

Originality. Originality in this investigation was scored using two main categories of methods: human raters and text-mining models. In addition, the reliability and validity of scores produced by a number of different types of text-mining models are compared. It should be noted here that, given critically important concerns about the way that any text-mining system deals with common function words (Forthmann et al., 2019), all of the analysis in this study utilizes inverse-document-frequency (IDF) term-weighting (Robertson & Jones, 1976) corrections to deal with extremely common words (e.g., "is"). Despite differences in how they were developed, each scoring system provides a model of language in a linear space, aiming for comparable distances between words in English. To score from each system's model, a weighted sum of word vectors is taken to represent each phrase for a response, and the cosine distance is taken between the response and AUT prompt (see Figure 1).

Human raters. First, every generated response from the 92 study participants across the 10 items on the AUT was coded for Originality by four human coders. In all, 5,491 responses were generated to the AUT in this study, with an average of 55.81 (SD = 31.72) per participant. The first Originality coder was the third author of this article, and the other three were paid research assistants. Each coder was instructed to score each generated response from 0–4, with zero being "totally ordinary" and four being "maximally novel." Coders were specifically trained to conceptualize most responses as being likely to fall toward the middle of that 5-point Originality scale: a belief that reflects our assumption that the Originality of generated responses is based on a continuous distribution, manifesting such that most responses are in the middle of the Originality scale (i.e., 2) with a smaller number of responses being completely unoriginal (i.e., 0) or very high on the Originality scale (i.e., 4). Such a continuous distribution of Originality—rather than discrete Originality categories—has been empirically documented in the literature (e.g., Dumas, 2018).

The four human coders coded the 5,491 responses with a "fair" level of interrater agreement (Fleiss' κ = 0.2198; Fleiss & Cohen, 1973). Typically, within the psychological research literature, any lack of exact agreement among coders would be resolved through discussion until all coders were able to converge on an agreed-upon categorical rating for every response (Gwet, 2014). Such a method operates under the measurement assumption that there is a true Originality category for each generated response (i.e., the latent Originality attribute is ordinal), and therefore coders must work to sort the generated responses into their true categories. However, an alternative method would assume that the 0–4 Originality categories the coders used were underlain by a continuous latent distribution, and therefore the originally coded Originality categories are meant to indicate locations on that continuous latent distribution.

Common in crowdsourcing methodology (e.g., Organisciak, Teevan, Dumais, Miller, & Kalai, 2014; Snow, O'Connor, Jurafsky, & Ng, 2008), where varying judgments of quality from raters are regularly aggregated, this continuity assumption suggests that exact categorical agreement among raters is not crucial, because, over multiple raters, a consensus about where on the underlying Originality distribution each generated response may be located can arise through averaging the ordinal category codes across raters. The intuition here is that disagreement among raters on ordinal codes is actually instructive and valuable to researchers. For example, if three raters coded a particular AUT response as a '3' on the Originality scale, but one rater coded it as a '4,' that last rating is still considered a nudge toward the more novel end of the scale for that response, and an averaged Originality score for that response of 3.25 would be considered closer to "true" than simply the modal rating of 3. For this reason, we created an aggregated human-coded Originality rating for each AUT response by averaging each of the 4 coders' ratings for every one of the 5,491 responses. Then, to aggregate those response-level Originality ratings to the participant-level, we further averaged each of those response ratings within each AUT prompt (e.g., Book) for every participant. This procedure resulted in 10 prompt-level human-rated Originality scores for each of the 92 participants in the dataset. Because it is the main focus of this investigation, further analysis with these human-rated Originality scores (e.g., modeling an underlying latent Originality attribute across all AUT prompts) are included in the Results section of this article. Further, the issue of the interrater reliability of these human-rated Originality scores is returned to with a critical lens in the Discussion section.

Text-mining systems. Here, we systematically compare the capability of four different publicly available text-mining systems to create reliable and valid participant Originality scores on the AUT. Each of these four text-mining systems differ in a variety of ways, including the corpora of text that they are trained on, the parameterization and specification of the statistical models they use, and the way they correct for difficult-to-model aspects of real-world language use such as words with multiple meanings and synonyms. Generally, larger corpora will more accurately represent the relations between words in the language, though the

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domain of the documents will lead to differences in how the language is interpreted and may affect the transferability of that particular model. For example, is *bank* more associated with *river* or *money*? A naive algorithm learning English from a collection of documents will decide that answer differently based on what those documents were written about. The sizes and domains of the corpora on which each system was trained are noted in Table 1. Different systems may also correct for perceived importance of words, deemphasizing common function words (e.g., and, the) or removing them altogether. Finally, all the system models are trained using different training methods. These methods differ on choices such as what frame of *document* suggests a relationship between words and how the training algorithm implements that theory. The choice of how many latent dimensions are learned also affects the system: Too few dimensions will lack depth and discriminatory value between words, whereas too many will overfit to the documents. In the current investigation, we do not attempt to absolutely control for every possible methodological option in the training of a text-mining model. Instead, we focus on already created text-mining systems that creativity researcher are currently able to access free of charge, to provide a meaningful demonstration of the strengths and weaknesses of each extant system specifically in the context of creativity and divergent thinking research.

Each of the four text-mining systems that are tested in this study are succinctly explained below. A bulleted explanation of each of these text-mining systems also appears in Table 1. Analysis based on these text-mining systems was accomplished by remotely accessing their freely available systems via the Python programming language. All reproducible computational code used in this investigation are freely available online via our laboratory ongoing Github account (https://github.com/massivetexts), and a static depository of the code used for this study is also available on the Open Science Foundation (https://github.com/massivetexts). In addition, computational code is available as online supplemental materials published with this article.

Touchstone Applied Science Associates (TASA) LSA. This system, which is by far the most commonly applied in the extant literature on divergent thinking (e.g., Dumas & Dunbar, 2014; Forster & Dunbar, 2009; Forthmann et al., 2019), is trained on a corpus of 37,651 educational texts and was originally intended to mimic to expected reading experience of the typical entering undergraduate student. This system was used by Landauer and Dumais (1997) in their initial demonstration of the capability of LSA to approximate the human semantic relations, but since then has been outstripped by other systems in terms of corpus size and model sophistication (Crossley et al., 2017). For example, recent work in the information sciences has confirmed Landauer and Dumais' (1997) argument in showing that LSA spaces trained on TASA do tend to match human semantic judgments but has also found better performance with larger corpora (Ştefănescu, Banjade, & Rus, 2014). In this study, we use the publicly available TASA model trained by Günther et al. (2015).

English 100k LSA. Also originally trained by Günther, Dudschig, and Kaup, the English (EN) 100k LSA text-mining system was previously applied to divergent thinking task data by Forthmann and colleagues (2019). This system is trained on a concatenation of multiple general purpose corpora of texts: a Wikipedia image, the general text British National Corpus, and a web crawl corpus that together included more than 5 million documents. After an initial modeling of the language in these 5 million documents, the 100,000 most frequently occurring unique words were retained to build the system (hence the 100k in the name). So, although the LSA training method in this system is the same as that in the TASA system, the size and generality of the corpora used in this the EN 100k system may be more advantageous for the quantification of originality on DT tasks because the more general corpora on which this model is trained may better represent the true semantic space from which DT task participants draw their responses.

Global Vectors for Word Representation 840B. Publicly available through the Stanford natural-language-processing laboratory (Pennington et al., 2014), but never before applied to the analysis of divergent thinking task data, the Global Vectors for Word Representation (GloVe) 840B text-mining system was trained on a corpus of 840 billion words that were scraped from a variety of online sources including Wikipedia and Twitter. Although GloVe is similar to the LSA-based text-mining systems in that its goal is to quantify the semantic relation between two words or phrases within a geometric space, GloVe accomplishes this goal through a probabilistic modeling framework. In addition, GloVe calculates correlations among terms in a more targeted way than does LSA: by examining a small window of word co-occurrence around each term where it is used, rather than examining co-occurrence in full-text documents. This shift in the mathematical and statistical underpinnings of the text-mining systems may hold potentially positive impact on the measurement of AUT originality in that it may potentially produce more stable and reliable estimates of response Originality (and this hypothesis will be tested in the current study).

Table 1
Short Description of Text-Mining Systems Included in This Investigation

System name	Training corpora	Training scale	Reference
TASA LSA	Multi-subject educational texts	37.7 thousand documents (92.4 thousand unique words)	Landauer & Dumais, 1997
EN 100k LSA	Wikipedia, ukWaC (web crawl), and British National Corpus (general)	5.4 million documents (2 billion words, 100 thousand unique words)	Günther, Dudschig, & Kaup, 2015
GloVe 840B	Common Crawl (web documents from sites including Wikipedia and Twitter)	840 billion words (2.2 million unique words)	Pennington, Socher, & Manning, 2014
Word2Vec	Google News (articles)	100 billion words (3 million unique words)	Mikolov, Sutskever, et al., 2013

Note. TASA = Touchstone Applied Science Associates; LSA = Latent Semantic Analysis; EN = English; GloVe = Global Vectors for Word Representation; Word2Vec = word-to-vector.

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Word2Vec. Named for the "word-to-vector" methodology it employs, Word2Vec focuses specifically on word-level corpora scraped from massive online sources of text (Mikolov, Chen, et al., 2013). This text-mining system was created at, and is publicly available through, the tech company Google, and was trained on a corpus of 100 billion words scraped from the news-aggregator Google News. Word2Vec modeling methodology focuses on the context of individual words, and through a neural network predictive modeling approach, works to predict a target word from a sample of closely co-occurring words (i.e., context words). In Word2Vec parlance, this method is termed "skip-gram," because the model skips individual target-words when training and then predicts the skipped word based on the context-words that co-occur with it. Previous research has found that this Wor2Vec method preserves the true semantic relations among words more effectively than other training models such as LSA (Mikolov, Sutskever, Chen, Corrado, & Dean, 2013). Further, Word2Vec models perform well at identifying high-order analogical relations among words (Bianchi & Palmonari, 2017). Because of the cognitive similarity between analogical and divergent thinking (e.g., Green et al., 2012), this finding may suggest that Word2Vec methods are particularly suited for the measurement of originality in DT tasks.

Results

The analysis for this psychometric investigation of text-mining-model-based Originality scoring systems unfolded in the following stages: (a) a careful investigation of the reliability of participant Originality scores generated by both human raters and text-mining systems, with an eye toward both composite and latent factor reliability; (b) an analysis of the correlations among human rated and text-mining system generated Originality scores; and (c) a criterion validity analysis of Originality scores in which the correlations from both human rated and text-mining system generated Originality to Fluency, Elaboration, Openness, Intellect, and Creative Activities were examined. Each of these three analytic stages are explained, and results are presented, below.

Reliability

Here, reliability of each of the five included Originality scoring methods (i.e., human raters, TASA LSA, EN 100k LSA, GloVe 840B, and Word2Vec) is examined using both observed variable (i.e., Classical Test Theory [CTT]) and latent variable (i.e., Confirmatory Factor Analysis [CFA]) methods.

Composite internal consistency. Human-coded Originality ratings on the 10 AUT prompts displayed a high level of composite or scale reliability (see Table 2 for reliability coefficients). In contrast, the composite reliability of the text-mining system generated Originality was substantially lower, although the TASA LSA and GloVe 840B each reached levels of composite reliability that may be generally considered acceptable in the psychological research literature (i.e., .80 or above). These coefficients are especially important given the propensity of creativity researchers for using summed-scores, rather than optimally weighted latent variable scores, in their research. However, a more in-depth analysis of the measurement properties of the Originality scoring systems is necessary to understand the way these scores relate to a latent Originality construct.

Table 2
Composite and Factor Reliability Indices for Originality Scores From Each Scoring System

Scoring system	α	ω	*H*
Human raters	.943	.948	.952
TASA LSA	.813	.825	.867
EN 100k LSA	.730	.758	.825
GloVe 840B	.800	.807	.875
Word2Vec	.741	.743	.807

Note. TASA = Touchstone Applied Science Associates; LSA = Latent Semantic Analysis; EN = English; GloVe = Global Vectors for Word Representation; Word2Vec = word-to-vector.

Confirmatory factor analysis. A unidimensional CFA model, in which all 10 AUT prompts loaded on a latent Originality factor, was fit to item-scores generated by each of the five Originality scoring systems (please see Figure 2 for a conceptual path diagram of this CFA model). Theoretically, such a model corresponds to a measurement assumption that all the administered AUT prompts (when scored for Originality) indicate a single underlying originality construct and therefore represents common measurement practice in the creativity literature (e.g., Storme et al., 2017). These CFA models were fit using maximum likelihood estimation in Mplus Version 8.0 (Muthén & Muthén, 2019). Based on the model root mean square error of approximation (RMSEA; See Table 3 for exact values), none of these unidimensional methods achieved a level of model-data fit that would be considered ideal in the methodological literature (i.e., below .06; Hu & Bentler, 1999; McNeish, An, & Hancock, 2018). However, the models for both the human raters and the GloVe text-mining system achieved a level of fit that would meet current standards in the creativity literature, where measurement model-data fit is often slightly weaker than in more traditional measurement areas such as reading or math (e.g., Yoon, 2017).

In addition, although the scoring systems differed in the strength of their CFA loadings (see Table 3 for loading details), the individual AUT items also displayed general trends in the strength of their loadings across scoring systems. For example, the prompt *Rope* displayed weaker standardized loadings than other prompts across multiple of the scoring systems, whereas the prompt *Bottle* displayed stronger loadings across multiple scoring systems. This pattern may likely be attributable to differential participant familiarity with certain objects, or perhaps the actual functional capabilities of each object to facilitate original alternate uses. One anomaly in these general patterns were the extremely weak standardized loadings for *Book* and *Table* in the EN 100k LSA system, and the model-data fit or this scoring system was also poor compared with the other models, so that may indicate that this corpus does not well-represent the true semantic relations among *Book*, *Table*, and their associated uses.

Many-faceted Rasch analysis for human raters. The confirmatory factor modeling perspective above intentionally aggregated human rated originality across all the responses to a given AUT prompt (e.g., Book), for all four of the human raters, through averaging. This method was designed to treat the raters as equally weighted voters in terms of the originality of a given AUT response, and therefore also for participant originality scores. How-

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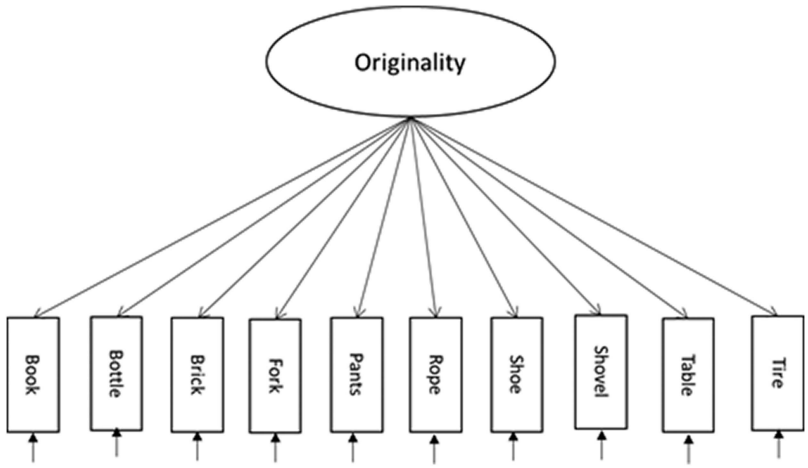

Figure 2. Conceptual path diagram of the latent measurement model used to determine the reliability of the Alternate Uses Task (AUT) scoring methods.

ever, a reasonable alternative perspective, recently demonstrated by Primi and colleagues (Primi, Silvia, Jauk, & Benedek, 2019) would be to employ a Many-Facet Rasch Model (MFRM) to incorporate differences in the leniency or severity of individual human judges into the calculation of originality scores. The MFRM model has been previously demonstrated to be useful in creativity research, in particular in modeling measurement error associated with multiple human raters (Barbot, Tan, Randi, Santa-Donato, & Grigorenko, 2012), and this previous usage suggests relevance of this modeling tool to the current study. So, as a further point of comparison to the CFA approach presented above, an MFRM with three facets (i.e., judges, items, and participants) was fit to these data using the specialized computer program Facets (Linacre & Wright, 1988), and used to calculate originality scores for each participant in the dataset. It should be noted that the use of the term "facets" is different in this context than in the personality measurement context. In the personality measures used in this study, the facets refer to the finely grained subscales within the Big 5 factors. In MFRM, a facet refers to a source of measurement error, in this case error may arise from inconsistencies among the human raters, AUT items, or the distribution of participant original thinking ability.

To accommodate the MFRM here, modal ratings for each judge were utilized for each AUT prompt (as opposed to means within each prompt as was used for the CFA above). MFRM parameter estimates are presented here in Table 4, which contain the difficulty/severity for the AUT items and raters, as well as the parameter-theta (latent score) correlations. As can be seen in this table, some AUT items were more difficult for participants to think of original uses for (e.g., Shoe; difficulty = .60), whereas other items were easier (e.g., Brick; difficulty = −.53). Similarly, some human raters were more lenient in judging originality of responses (e.g., Rater 3; severity = 1.4), whereas others were more severe (e.g., Rater 2 = −.66). As a general measure of internal consistency, the Rasch average reliability among the 10 AUT items in the MFRM was .90, and therefore MFRM based scores were generated and saved in our dataset for future analysis. As an alternative to MFRM not applied here, interested readers should also see the application of item-response models to data from multiple human-raters recently posited by Myszkowski and Storme (2019).

Latent factor internal consistency. Using the standardized loadings and residual variances that are generated when fitting the CFA models, we then calculated two modern factor reliability statistics for each of the scoring systems: Omega (McDonald, 1999) and *H* (Hancock, 2001). Although both of these indices represent a more sophisticated estimate of the score reliability than Cronbach's alpha, they differ in their assumptions about the way participant scores will be produced in future investigations using a

Table 3
Confirmatory Factor Model Parameters for Each Scoring System

		Alternate uses task prompt standardized loadings									
Scoring system	Model RMSEA	Book	Bottle	Brick	Fork	Pants	Rope	Shoe	Shovel	Table	Tire
Human raters	.063	.837	.867	.739	.798	.864	.642	.788	.807	.825	.838
TASA LSA	.082	.732	.807	.655	.429	.414	.433	.744	.327	.497	.562
EN 100k LSA	.124	.023	.647	.513	.433	.650	.629	.742	.499	.039	.564
GloVe 840B	.069	.672	.784	.805	.595	.590	.161	.752	.268	.309	.350
Word2Vec	.079	.605	.761	.488	.100	.469	.336	.686	.457	.421	.334

Note. TASA = Touchstone Applied Science Associates; LSA = Latent Semantic Analysis; EN = English; GloVe = Global Vectors for Word Representation; Word2Vec = word-to-vector; RMSEA = root mean square error of approximation. All standardized loadings significant at $p < .05$ except Book and Table in the EN 100k LSA system.

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Table 4
Many Facet Rasch Model Parameters for Human Rated AUT Items

Model parameter	Difficulty/Severity (*SE*)	Parameter-Theta correlations
AUT Items		
Book	.13 (.09)	.71
Bottle	.12 (.09)	.78
Brick	−.53 (.10)	.70
Fork	−.06 (.09)	.71
Pants	−.15 (.09)	.81
Rope	.05 (.09)	.62
Shoe	.60 (.09)	.74
Shovel	−.35 (.10)	.71
Table	.11 (.09)	.76
Tire	.09 (.09)	.76
Human raters		
Rater 1	−.63 (.06)	.70
Rater 2	−.66 (.06)	.69
Rater 3	1.4 (.06)	.71
Rater 4	−.11 (.06)	.70

Note. AUT = Alternate Uses Task; *SE* = standard error.

measure. If, in the future, Originality scores are created by summing or averaging across multiple AUT prompts, then Omega is the best representation of the reliability of those scores, but *H* assumes an optimally weighted measurement model in which the Originality scores are estimated directly from a CFA or item-response model (McNeish, 2018). For this reason, Omega is always slightly lower than *H* to account for the added measurement error associated with summing or averaging scores across a measure rather than using a psychometric model.

As can be seen in Table 2, the human-raters achieved by far the most reliable Originality scores. However, all of the text-mining systems, at least in terms of coefficient *H*, achieved an acceptable level of factor reliability (i.e., above .80) as well, although the most reliable system (i.e., GloVe) was substantially more so than the least (i.e., Word2Vec). This finding implies that, should researchers generate scores from a latent measurement model, all of the scoring systems included here are capable of producing generally acceptable scores (although GloVe scores would be the most reliable, and have the best model-data-fit). In terms of Omega—which converged on the same results as Cronbach's alpha—only the TASA LSA system and GloVe achieved acceptable reliability, indicating that these are the only systems that produce stable enough scores to warrant calculating a simple composite score (e.g., a sum) as opposed to using a latent scoring model. Going forward, Originality scores for all 92 participants from each of the five scoring system were generated via empirical Bayes using the SAVEDATA command in Mplus.

Relation Between Text-Mining Models and Human Raters

Perhaps the most critical test of the efficacy of the text-mining scoring systems is their ability to approximate the Originality scores produced by human raters. Table 5 holds the correlations among the five scoring systems included in this investigation. As expected, the human rated Originality scores generated via CFA and MFRM were correlated very strongly (.98). More critical are the correlations among the text-mining systems and the human raters. These correlations show that the GloVe text-mining system is the most capable of producing Originality scores that resemble those of humans. In contrast, the Word2Vec system's Originality scores were the most weakly correlated with human-rated scores. Although, it should be noted, all four of the systems utilized here produced latent Originality scores that were significantly and positively correlated with scores from human raters. In addition, the Originality scores from each of the four text-mining systems correlated strongly (in the .9's) with one another, indicating that—although the systems differed in their observed and latent reliability indices and CFA model-data fit—overlapping information about participant Originality is provided by each system.

Criterion Validity

Here, the capacity of the five Originality scoring systems to produce scores that predict other common indicators of creativity (i.e., Fluency, Elaboration, Openness, Intellect, and Creative Activities) are systematically examined. See Table 6 for correlations discussed in this section.

Fluency. The theoretical relation between Fluency and Originality is currently debated in the creativity research literature, with some scholars arguing for a positive, zero, or negative correlation among these dimensions of divergent thinking (see Dumas & Dunbar, 2014 or Forthmann et al., 2019 for discussions of this issue). Here, human-rated Originality scores (calculated via CFA) correlated weakly and positively (and nonsignificantly) with Fluency scores, implying that that—should we take the human-rated scores as baseline truth—the actual relation between these dimensions is close to zero, at least in this general-population sample. However, all of the text-mining systems in this study displayed

Table 5
Correlation Matrix Among Originality Scores From All Scoring Systems Included in This Investigation

Scoring system	Human raters (CFA)	Human raters (MFRM)	TASA LSA	EN 100k LSA	GloVe 840B	Word2Vec
Human raters (CFA)	1.00					
Human raters (MFRM)	.98	1.00				
TASA LSA	.67	.56	1.00			
EN 100k LSA	.66	.55	.93	1.00		
GloVe 840B	.73	.63	.97	.96	1.00	
Word2Vec	.58	.45	.96	.95	.94	1.00

Note. TASA = Touchstone Applied Science Associates; LSA = Latent Semantic Analysis; EN = English; GloVe = Global Vectors for Word Representation; Word2Vec = word-to-vector; CFA = confirmatory factor analysis; MFRM = Many-Facet Rasch Model. All correlations significant at $p < .01$.

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Table 6
Correlations Among Originality Scoring Systems and External Criteria

Scoring system	Ideational fluency	Elaboration	Openness	Intellect	Creative activities composite
Human raters (CFA)	.176	.215*	.088	−.086	−.093
Human raters (MFRM)	.103	.186	.039	−.117	−.100
TASA LSA	.346**	.239*	.119	.010	−.077
EN 100k LSA	.264**	.273**	.046	.044	−.083
GloVe 840B	.336**	.233*	.084	−.018	−.077
Word2Vec	.365**	.339**	.084	.014	−.025

Note. CFA = confirmatory factor analysis; MFRM = Many-Facet Rasch Model.
* $p < .05$. ** $p < .01$.

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significant and positive correlations (although only moderate in strength) with Fluency. Given this finding, it appears that human-rated Originality scores have the greatest degree of discriminant validity from Fluency scores, whereas text-mining models under investigation produced Originality scores that were much more strongly associated with Fluency. In particular, the EN 100k system displayed the lowest correlation to Fluency among the text-mining systems, making it the most consistent with human-rated Originality scores in that regard.

Elaboration. In previous work with text-mining system Originality scores (Forthmann et al., 2019), the relation between Elaboration and Originality has been considered a source of bias in the scores. In this investigation, following previous methodological recommendations, the IDF correction for stop-words was utilized. Here, we found that the human raters' Originality scores (calculated via CFA) were significantly and positively associated with Elaboration (which is a stronger relation than those human-rated scores had with Fluency). In contrast, the correlation between MFRM calculated Originality ratings and elaboration was not significant (i.e., $p = .072$). Following the pattern set by the CFA produced Originality ratings, all of the text-mining systems also produced scores that were significantly and positively associated with Elaboration, although all of the text-mining systems displayed correlations to Elaboration that were stronger than that of the human-raters: a findings that highlights previously observed discriminant validity issues, even with the IDF correction. However, given that the human rated scores were also positively associated with Elaboration, it appears that the text-mining systems are not much more confounded with Elaboration than are human raters, although substantial variation among the text-mining systems was observed. Specifically, the GloVe system was most capable of preserving the low-moderate correlation with Elaboration, followed by TASA. The Word2Vec system produced the strongest correlation with Elaboration.

Openness and intellect. In this study, none of the Originality scoring systems (human-rated or text-mining) produced scores that significantly correlated with Openness or Intellect. However, in the case of both of these creative-personality indicators, the GloVe system produced correlations that were closest to those of the human raters, indicating that the GloVe originality scores approximated human-rated Originality scores the best in regards to their relation to creative personality variables.

Creative activities. When predicting the composite of the Creative Activities measure, none of the Originality scoring systems produced significant correlations, although again the GloVe system was most in-step with the human-raters. At the more fine-grained level of the individual scales of the Creative Activities measure (see Table 7), the human-rated Originality scores (calculated either by CFA or by MFRM) did significantly but negatively correlate with creative Cooking activities. These findings imply that, at least in this general-population sample, AUT Originality scores are not related to the self-reported domain-specific creative

Table 7
Correlations Among Creativity Indicators and Creative Activities

Creativity indicator	Performance	Arts	Cooking	Crafts	Literary	Music
Originality scoring systems						
Human raters (CFA)	−.120	−.081	−.227*	−.075	.108	.029
Human raters (MFRM)	−.119	−.097	−.243*	−.081	.093	.044
TASA LSA	−.156	−.022	−.139	−.022	.108	.025
EN 100k LSA	−.155	−.041	−.103	−.051	.085	.033
GloVe 840B	−.134	−.038	−.152	−.063	.107	.053
Word2Vec	−.116	.016	−.085	.042	.134	.033
Creative personality indicators						
Openness	.134	.307**	.266*	.392**	.309**	.201
Intellect	.052	.069	.465**	.339**	.133	.051
Divergent thinking dimensions						
Fluency	.150	.168	.147	.171	.309**	.254**
Elaboration	.080	.155	.029	.191	.220	.145

Note. TASA = Touchstone Applied Science Associates; LSA = Latent Semantic Analysis; EN = English; GloVe = Global Vectors for Word Representation; Word2Vec = word-to-vector; CFA = confirmatory factor analysis; MFRM = Many-Facet Rasch Model.
* $p < .05$. ** $p < .01$.

activities of participants. Among creative personality indicators, Openness significantly and positively predicted Arts, Cooking, Crafts, and Literary activities, whereas Intellect predicted Cooking and Craft activities. Ideational Fluency significantly predicted both Literary and Musical activities, but Elaboration did not significantly predict any of the included creative activities. Interestingly, none of these creativity indicators were capable of significantly predicting Performance activities, although the low-prevalence of Performance within this general sample (as opposed to a professionally creative sample) likely limited the variance of this activity scale and therefore precluded a significant correlation.

Discussion

As far as we are aware, this investigation has been the first within the creativity research literature to compare the ability of multiple text-mining models to produce reliable and valid Originality scores on the AUT, as compared with human-raters. As such, this study has a number of principal findings and specific recommendations to forward to the creativity research community. Four of these principal findings are described in detail below.

Human Raters Can Produce the Most Reliable Originality Scores

Within the creativity literature, the perennially low level of exact agreement (and therefore low level of interrater reliability) between human raters on the level of Originality of a given AUT response has led many to bemoan the possibility of producing highly reliable creativity research using human raters (see Storme, Myszkowski, Çelik, & Lubart, 2014 for one approach to improving this reliability). However, our results show that, if researchers are willing to conceptualize human-rated Originality codes as ordinal indicators of an underlying Originality continuum and therefore average the Originality codes across raters, a very high level of reliability is possible with four trained raters. Of course, this high level of reliability refers to the consistency of the continuous Originality scores that are created by aggregating all 10 of the AUT prompts included in this study (either by summing or through a CFA), and not to the individual AUT responses that were coded separately by each human rater. If an analysis at the fine-grained level of an individual response to a specific AUT prompt was desired by a researcher, then interrater reliability may be a more informative index. Although such fine-grained analysis at the individual response level is somewhat common among creativity researchers with a basic psychology focus (Benedek, 2018), those researchers whose work is more situated within an applied psychology area (e.g., educational psychology; Kerr & Stull, 2019) are more commonly concerned with the capacity of creativity measures to produce reliable and valid scores for participant-level interpretation across multiple items, tasks, or prompts. Following with that applied-psychology focus, should a researcher or practitioner desire to use the AUT to produce participant Originality scores for admission into a specialized educational program or personnel selection in the workplace, it does appear from these results that four trained raters can produce highly reliable Originality scores across 10 AUT prompts, at least with this general-population adult sample. Of course, the high level of reliability achieved here by the human-raters was likely influenced, at least in part, by the specific training and feedback that we provided our raters. In this case, raters were specifically trained to conceptualize the Originality scale along which they rated AUT responses (that ranged from 0 to 4) as a continuous dimension, on which most responses would have a moderate amount of Originality (i.e., a 1, 2, or 3), and only a few responses would fall on the extremes of the scale (i.e., 0 or 4). Using this particular prompting, the participant-level latent Originality scores we calculated were able to achieve a high level of reliability. In future work, it should not be assumed a priori that the highest score reliabilities are possible with human raters as opposed to text-mining systems. In cases where the human raters are not effectively trained or are less motivated, the scores from human raters could actually be less reliable than those from text-mining systems.

GloVe 840B Is the Recommended Text-Mining System for Originality

A major stated goal of this investigation was to identify the publicly available text-mining system that is most capable of producing reliable and valid Originality scores on the AUT. Across the stages of the current investigation, the GloVe 840B system emerged as the best system to choose for producing Originality scores in creativity research. The GloVe system generated the most reliable scores within a CFA framework as indicated by coefficient *H,* and the second most reliable composite scores as indicated by Alpha and Omega. The only system that produced more reliable composite scores (i.e., TASA LSA) demonstrated substantially worse model-data-fit of the unidimensional CFA model as indicated by RMSEA. In addition, the TASA LSA scores correlated substantially weaker with human-rated Originality than did the GloVe scores, a strong indication that GloVe scores are more valid than TASA scores. In addition, although both TASA and GloVe scores did display the previously described potentially problematic relation to Elaboration (Forthmann et al., 2019), so too did the Originality scores coded by human-raters. The GloVe scores displayed the correlation with Elaboration that was most in line with the human-raters (although the difference between GloVe and TASA in this respect was not great). Further, among the text-mining systems, the GloVe scores had the correlations to other creative indicators (e.g., Openness and Intellect) that were most similar to that of human raters, although it should be noted that none of the Originality scoring systems used here (humans or text-mining) was significantly correlated with these indicators of creative activity, with the exception of a weak-moderate and negative correlation between the human-rated Originality scores and Cooking activities. Such a general lack of covariance in this regard may be caused by the psychological differences in creativity-related self-report variables and more objectively quantified DT performance tasks such as the AUT. So, the near-zero correlations from GloVe-based Originality scores to the Openness and Intellect measures are here interpreted as a positive finding related to the validity of GloVe-based Originality scoring: GloVe was capable of producing Originality scores that generally mimicked the criteria correlations of the human-rated Originality scores, but much more quickly and at a greatly reduced cost.

In past investigations of text-mining models to produce Originality scores, by far the most commonly utilized text-mining

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system has been the TASA system (e.g., Dumas, 2018), with a small minority of other pieces utilizing the EN100k LSA system (Forthmann et al., 2019). Given the results of this investigation, researchers in this area should likely pivot their methodological focus away from LSA based systems to GloVe, to create more reliable and valid scores for research. GloVe's improved reliability and validity is likely attributable simultaneously to three factors: the size of its training corpus, the domain-generality of its training corpus, and its probabilistic modeling approach. In general, such a massive corpus (840 billion words, with 2.2 million unique words) may be more capable of approximating the actual semantic structure of language-use than a smaller corpus (e.g., TASA's unique words are only 4% of GloVe's), leading to more psychologically reliable and valid Originality scores. Further, the inclusion of text from sources like Wikipedia, a general web crawl, and the British National Corpus make GloVe much more general in scope than the TASA corpus that is composed of only educational texts. Finally, although both GloVe and LSA seek to represent the euclidean distance (or cosine similarity) between word vectors, LSA estimates these word vectors using a traditional parametric approach and GloVe uses a more modern log-linear, or probabilistic method that may produce more psychologically relevant results (Pennington et al., 2014).

Of course, all of the text-mining models compared here (i.e., TASA LSA, EN100k, word2vec, GloVe) are members of a larger family of dimension-reduction–based techniques for the quantification of semantic relations among words and phrases through the examination of the angles among estimated word vectors. As previously reviewed, an alternative approach would be to quantify the semantic distance among words using a network science approach (De Deyne et al., 2016), in which the semantic distance is not operationalized based on vector angles, but instead based on the path length between two or more words in the network. Some evidence in cognitive psychology suggests that the network science approach may have advantages above the dimension-reduction approach when studying fine-grain cognitive processes (e.g., priming; Kumar et al., 2019). However, it remains a future direction to ascertain whether the network science approach could be helpful in psychometric work that, as with the current study, aims to quantify Originality at the participant level with reliable and valid scores. Recent arguments in the creativity literature (Kenett, 2019) suggest that the network science approach to quantifying semantic distance may be fruitfully applied to computational psychometrics of creativity, and this approach may have promise for creativity researchers.

Not All AUT Prompts Contribute Equally to Reliability

One interesting, and perhaps problematic, aspect of creativity research is that the field's most commonly used measure (i.e., the AUT) is not necessarily fully standardized across studies in its administration procedures or even the particular prompts that are included. In this study, the particular measure-administration choices, as well as the particular prompts included on the AUT, may mean that the results could differ from other investigations where different measurement procedures were used. In addition, recent research (i.e., Beaty, Kenett, Hass, & Schacter, 2019) has shown that certain prompts may be more facilitative of Ideational Fluency or Originality, depending on the semantic richness of the ideas associated with that prompt. In essence, the findings of any psychometric research are tied to the actual item-stimuli that is administered to participants. Therefore, the findings of the current study are most relevant for those researchers who administer the same or similar AUT prompts as we administered here, and those researchers who plan to administer AUT prompts that are highly different than those administered here may need to interpret our results with caution.

However, this lack of standardization of the AUT may also be an unexpected boon for the creativity literature, in that the AUT measure or administration procedures can be easily updated as more psychometric evidence becomes available. In this study, we found substantial differences in the way individual AUT prompts (e.g., Book) contributed to the reliability of the latent Originality factor. For example, for both the human-rated and GloVe Originality scores, "Rope" was the weakest loading item, indicating that item detracts from the overall reliability of the Originality scores. However, this effect was more pronounced in GloVe than for the humans, implying that the human raters were capable of providing relatively stable estimates of participant responses for "Rope," whereas GloVe struggled more to quantify the relevant semantic relations for that prompt. In addition, the loading for "Shovel" on the GloVe Originality factor was also relatively weak, implying that text-mining system is worse at representing the semantic relations around "Shovel" than it is for the semantic relations around "Brick," for example. Overall, it is clear from these findings that the text-mining–based scoring methods had much more varying loadings across the AUT prompt than did the human-raters, which illustrates the need, for those researchers who use text-mining systems to measure Originality, to choose their AUT prompts carefully, and possibly pilot them with their chosen text-mining system before administering them to a large number of participants.

Multiple Dimensions of Creativity Are Needed for Research

Although this current investigation was mainly focused on the psychometric quality of Originality scores, the validity portion of the study also offers some valuable insight into the interrelations among other dimensions of DT (i.e., Fluency and Elaboration), creative personality (i.e., Openness and Intellect), and particular real-world creative activities. Although the AUT-based Originality scores did not well-predict self-reported creative activities, other dimensions of the AUT scoring (i.e., Fluency and Elaboration) did. For example, the capability of Fluency scores to significantly and positively predict literary and musical creative activities, and the prediction of arts activities with AUT Elaboration, highlight the continued usefulness of both AUT Fluency and Elaboration scores in the creativity literature. However, creative personality indicators were even better able to predict self-reported creative activities, with Openness being the most predictive (i.e., significant positive correlations with Arts, Cooking, Crafts, and Literary Activities) and Intellect also being reasonably predictive (i.e., significant positive correlations with Cooking and Crafts activities). Of course, given the self-report nature of the Creative Activities Inventory as well as the personality questionnaires, their relations may be inflated by participants' creative self-concepts (Karwowski, 2016). But, such an explanation can be ruled out for Fluency and Elaboration, both of which predicted creative activities as well as or stronger than human-rated Originality. Perhaps most importantly, it should be observed that, of the creative activities that

were significantly predicted in this study, none was predicted by all of the creativity indicators. This finding strongly highlights the need for researchers to measure a variety of different indicators of creative potential—including both DT and personality assessments—to maximize the impact of our research to understand the creative process. As a future direction in this line of investigation, it may be important for researchers interested in the psychometrics of creativity to consider ways not only to automate Originality scoring using text-mining or other computational methods, but also to automate scoring systems for other dimensions of divergent thinking and creative potential. For example, it remains to be seen whether or how text-mining systems can be used for the quantification of Flexibility on the AUT or other DT measures. In addition, creativity researchers have been perennially interested in participant responses to the AUT that stand out because of their sexual or violent content (Dumas & Strickland, 2018; Hudson, 1968), and text-mining models could conceivably be applied to automatically identify such responses. In our view, these future directions help illustrate the potential of computational psychometric work in the creativity research area.

Coda

In our view, one main methodological bottleneck that limits the productivity and impact of creativity research has historically been the time- and resource-intensiveness of human-rated DT tasks. We in the field have relied on hiring, training, and compensating human-raters for decades, and graduate students situated within creativity research laboratories have also often shouldered the burden of rating hundreds or thousands of DT responses for Originality. In some cases, very large–scale studies of creativity may even have seemed infeasible because of the burden of using human-raters. Here, we found the GloVe system is highly capable of approximating Originality scores produced via human-raters, but much more rapidly and potentially free of cost. Based on the findings of this study, we may be nearing a time in the field when the work of scoring DT tasks for Originality can be automated using a text-mining model, opening the door for much larger-scale studies of DT and creativity, and hopefully leading to increased reach and scope of research in the field. In the future, such text-mining systems may be even easier to run (e.g., through more user-friendly software) and may contribute to a streamlined and automatic process of Originality measurement in creativity research.

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Received June 3, 2019
Revision received January 29, 2020
Accepted February 19, 2020 ■

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